

Kevin Prakash J — Portfolio Master Profile & Creative Direction

Verified master source for portfolio content, positioning, creative direction, TeamFlow scope, and implementation planning.

1. Personal & Academic Profile

- Name: Kevin Prakash J.
- B.Tech Computer Science & Engineering at VIT University, Vellore.
- Academic timeline: August 2023 – May 2027.
- Current stage: Fourth-year undergraduate student.
- Current CGPA: 8.72 / 10.
- Primary professional interests: Artificial Intelligence, Machine Learning, NLP, Computer Vision, LLMs, multimodal AI, software engineering, and applied research.

2. Professional Identity

- Kevin is an AI/ML-focused software engineer and researcher who builds practical intelligent systems.
- The portfolio should communicate three connected strengths: AI and research, computer vision, and software engineering.
- The positioning should emphasize real systems, research depth, technical problem-solving, and strong interaction design rather than presenting a conventional student resume.
- Desired impression: technically strong, research-aware, experimental, practical, and capable of turning real-world problems into usable systems.

3. Technical Skills

- Programming: Python, Java, C, C++.
- Web and software: HTML, CSS, JavaScript, React, Flask, Tailwind CSS, ASP.NET Core, TypeScript, C#, and REST APIs.
- AI/ML: PyTorch, TensorFlow, Hugging Face, LangChain, Ollama, OpenCV, MediaPipe, prompt engineering, NLP, and LLMs.
- Databases: MySQL. SQLite has been used in earlier project work but is not a current resume focus.
- Testing: PyTest and mutation testing.
- Tools and platforms: Git, GitHub, Linux, VS Code, Streamlit, and PyInstaller.

4. Industry Experience

- AI Intern at Syncfusion Software Pvt. Ltd., May-June 2026.
- Technology exposure included ASP.NET Core, React, TypeScript, REST APIs, role-based access control (RBAC), C#, and Git.
- This experience adds industry software-engineering evidence alongside Kevin's research and personal projects.

5. Projects

- MyKitchen — Dish Recommendation Planner: a full-stack application for dish recommendation and meal planning, built around Python/Flask with a web interface and database-backed functionality.

- **AI-Based Email Summarizer:** an NLP/LLM application using Python, LangChain, and large language models to summarize email content.
- **AWS Attendance System:** a cloud-oriented attendance project built around AWS EC2, S3, and RDS.
- **Indian Sign Language Translator:** a computer-vision system using camera input, OpenCV, MediaPipe hand landmarks, gesture recognition, text generation, and speech output.
- **TeamFlow — Full-Stack Team & Project Management Platform:** a full-stack platform for centralized team, project, and task management. The portfolio treats the project as a finished product with authentication, team membership, role-based access control, project management, task workflows, Kanban-style task management, collaboration features, and a structured React/FastAPI/PostgreSQL architecture.

6. TeamFlow — Final Portfolio Project Profile

- TeamFlow is a full-stack team and project management platform built to centralize teams, projects, tasks, and collaboration.
- **Core stack:** React, TypeScript, Vite, React Router, Python, FastAPI, Uvicorn, SQLAlchemy 2.x, Pydantic, PostgreSQL 17.11, JWT, bcrypt, pytest, Git, and GitHub.
- **Authentication:** registration, login, JWT bearer authentication, bcrypt password hashing, current-user retrieval, logout handling, protected frontend routes, persistent authentication, and inactive-user enforcement.
- **Teams:** team creation, team listing, team details, team updates, team deletion, membership management, member listing, owner/admin/member roles, role changes, member removal, member departure, and ownership protection.
- **Project management:** team-scoped project creation and management with project status handling.
- **Task management:** task creation and assignment, todo/in-progress/review/done workflow, ordering, filtering, and Kanban-style task management.
- **Collaboration:** task comments, activity logs, and notifications.
- **Architecture:** React + TypeScript + Vite frontend; FastAPI backend organized into routers, schemas, services, SQLAlchemy models, database layer, and authentication/security dependencies; PostgreSQL database with User, Team, TeamMember, Project, and Task-oriented resources.
- **Authorization boundary:** team membership is the central authorization boundary for downstream team resources, with owner/admin/member permissions.
- **Testing and engineering:** automated backend tests, API verification, browser-level authentication verification, database integrity constraints, service-layer separation, protected React routing, and incremental phase-based development.
- **Important positioning:** TeamFlow is a software-engineering project, not an AI/ML project. The portfolio should emphasize architecture, authentication, authorization, database design, API design, workflows, and collaboration features.

7. Research — Multilingual NLP

- **Research title:** “Cross-Lingual Semantic Alignment and Language-Family Structure in Multilingual Sentence Embeddings for Indic Languages”.
- **Focus:** studying semantic alignment across Indic languages and how language-family structure is reflected in multilingual sentence embeddings.
- **Datasets/corpora discussed:** Samanantar, Wiki-Translate, and IN22/Indic-language resources.
- **Models considered:** mBERT, XLM-R, LaBSE, and MuRIL.
- **Evaluation directions:** semantic affinity, graph modularity, manual STS correlation, bitext retrieval evaluation, and language-family-aware analysis.

8. Research — One Health / Avian Influenza

- Major research direction: a Digital Twin / One Health platform for predictive food and health security, with the implementation focused on Avian Influenza surveillance and prediction.
- Study focus: Avian Influenza across representative regions using a 2020-present timeframe and real-world, dynamically updated datasets.
- Representative regions discussed: India, Vietnam, United States, and Netherlands, with a broader 3-5-region implementation considered.
- Core architecture: Multimodal Prediction Engine → One Health Knowledge Graph → Agentic AI Decision Engine → LLM Reasoning + GraphRAG → Explainability Layer → Decision Support System.
- Potential data sources include ERA5 for weather, MODIS for satellite observations, FAO EMPRES for outbreaks, WOAHA WAHIS for veterinary surveillance, WHO FluNet for influenza surveillance, eBird for wildlife observations, FAOSTAT for livestock, UN Comtrade for trade, GDELT for news, and WHO/WOAH/FAO for knowledge.
- Three central novelty directions: a multimodal One Health Knowledge Graph; agentic AI for decision intelligence beyond prediction; and explainable LLM reasoning that provides evidence and rationale rather than only a risk score.

9. Research & Technical Interests

- Multilingual NLP and semantic alignment.
- Large language models and generative AI.
- Agentic AI and decision intelligence.
- Knowledge graphs and GraphRAG.
- Explainable AI.
- Computer vision and gesture recognition.
- Multimodal AI.
- AI for public-health, agricultural, and One Health decision support.
- Data engineering and reproducible dataset workflows.

10. Engineering & Dataset Workflow Philosophy

- Python should perform the actual parsing, cleaning, mapping, aggregation, validation, and output work.
- Prefer one script per dataset task and one-shot execution where practical.
- Front-load complete specifications before execution.
- Validate within the same processing script and produce compact validation reports.
- Avoid unnecessary exploratory commands, repeated inspection/edit/run cycles, and redundant verification.
- Before substantial dataset processing, perform a simple local availability, file-size, and, when easy, row-count check. If the substantive dataset is missing, stop rather than spending agent or processing effort on an empty or placeholder output.

11. Portfolio Creative Direction

- Overall concept: a reactive Spider-Man-inspired portfolio in which the web is part of the information architecture, not merely decoration.
- Visual language: dark black base with red, blue, and white accents; Spider-Verse/comic influence without becoming a generic fan site.

- Goal: a highly designed developer portfolio inspired by Spider-Man, rather than a Spider-Man-themed resume.
- Motion should feel physical and responsive: web shooting, attachment, pulling, snapping, transitions, node expansion, and cursor-aware movement.

12. Portfolio Navigation Concept

- Navigation should behave like a web rather than a conventional static navbar.
- Planned information architecture: Home, About, Skills, Projects, Research, Experience, and Contact.
- Selecting a navigation node should be able to trigger a web-shooting transition that pulls the visitor into the selected page.
- The web should connect sections and make navigation part of the experience itself.

13. Portfolio Skills-Web Concept

- Skills should be represented as an interactive web/network instead of a conventional grid.
- Parent categories act as nodes. Hovering a parent category causes related skills to pop outward along web-like connections.
- Example: hovering AI / ML causes PyTorch, TensorFlow, and Hugging Face to emerge from the parent node.
- Other relationships can include NLP → LangChain / Hugging Face / LLMs; Computer Vision → OpenCV / MediaPipe; and Web Development → React / Flask / JavaScript.
- The interaction should communicate relationships between skills instead of simply listing technologies.

14. Inspiration & Interaction Research

- Miu Miu Immersive Bags is a reference for immersive scene/page transitions and experience-driven navigation.
- TRIONN is a reference for strong typography, structured storytelling, project presentation, and interaction-led presentation.
- Awwwards-style experiences will be used as a source of interaction patterns such as experimental navigation, scroll-driven transitions, cursor behavior, project showcases, typography animation, loading sequences, and micro-interactions.
- These references are for extracting interaction ideas, not copying visual designs. The final experience should translate those ideas into Kevin's original Spider-Man/web language.

15. Proposed Portfolio Pages

- Home: high-impact introduction, web environment, identity, and entry into the portfolio.
- About: personal and professional identity, interests, approach, and current direction.
- Skills: interactive skill web with parent/child relationships and expandable technology nodes.
- Projects: visual project experiences for TeamFlow, Indian Sign Language AI, One Health AI, MyKitchen, Email Summarizer, and AWS Attendance.
- Research: multilingual NLP research and One Health/Avian Influenza research presented as technical stories.
- Experience: Syncfusion internship with role, technologies, responsibilities, and relevant outcomes.
- Contact: a simple contact experience integrated into the overall visual system.

16. Project Presentation Philosophy

- Projects should be presented as experiences rather than flat cards whenever practical.
- A project flow can be: web node → project entry → problem → architecture → technologies → implementation → results → GitHub/demo.
- TeamFlow should receive a particularly strong software-engineering case study covering architecture, authentication, RBAC, database design, APIs, workflows, testing, and collaboration.

17. Motion & Interaction Direction

- Web shooting and attachment animations.
- Page pull/transition effects.
- Skill-node expansion and web growth.
- Cursor-responsive effects where useful.
- Scroll-driven reveals.
- Typography and section transitions.
- Project hover physics and micro-interactions.
- Loading sequence that establishes the web environment.
- Motion should include a sensible reduced-motion fallback.

18. Career Direction

- Target direction: AI/ML and software engineering roles with meaningful exposure to intelligent systems, NLP, computer vision, LLMs, or applied research.
- Preferred internship geography: Chennai first and Bangalore second.
- Long-term positioning should support both industry AI/software roles and research-oriented opportunities.

19. Evidence & Portfolio Assets to Collect

- GitHub repositories and live demos for projects.
- Research papers, drafts, datasets, experiment outputs, and diagrams where appropriate.
- Internship details and evidence that can be publicly shown.
- Certificates, awards, competitions, publications, or other verifiable achievements if applicable.
- Professional links such as GitHub and LinkedIn.
- Final resume and selected screenshots/videos of working projects.

20. Final Phase Plan

- Phase 0 — Identity & Direction: lock portfolio identity, audience, visual language, and Spider-Man/web metaphor.
- Phase 1 — Inspiration & Interaction Research: analyze Miu Miu, TRIONN, and selected Awwwards experiences.
- Phase 2 — Information Architecture: finalize pages, content hierarchy, and navigation flow.
- Phase 3 — Spider-Web Navigation: design the web-based navigation and page-transition behavior.
- Phase 4 — Home Experience: storyboard the landing sequence and hero interaction.
- Phase 5 — Interactive Skills Web: design parent/child skill relationships and hover expansion.
- Phase 6 — Projects: design project storytelling and technical presentation, with TeamFlow as a major software case study.
- Phase 7 — Research: design research storytelling and evidence presentation.

- Phase 8 — Experience: present internship experience.
- Phase 9 — About: establish personal voice and professional identity.
- Phase 10 — Motion & Micro-interactions: define web, cursor, scroll, and transition behaviors.
- Phase 11 — Visual Design System: finalize typography, colors, components, spacing, and responsive behavior.
- Phase 12 — Build: implement the website after content and design decisions are locked.
- Phase 13 — Testing & Polish: responsive testing, accessibility, performance, navigation, animation, and final refinement.

21. Current Portfolio Baseline

- Confirmed academic profile: fourth year, CGPA 8.72/10.
- Confirmed major project portfolio: TeamFlow, Indian Sign Language Translator, One Health/Avian Influenza AI work, MyKitchen, AI Email Summarizer, and AWS Attendance System.
- Confirmed industry experience: AI Internship at Syncfusion Software Pvt. Ltd.
- Confirmed research directions: multilingual Indic-language sentence embeddings and One Health/Avian Influenza surveillance/prediction.
- Confirmed creative direction: reactive Spider-Man-inspired portfolio, web-based navigation, interactive skill web, and premium interaction references.
- TeamFlow is treated as a finished portfolio project for planning purposes, with its full intended feature set represented in the portfolio case study so the portfolio does not need a structural rewrite later. The exact implementation status can be validated separately if needed, but the portfolio information architecture is now designed around the complete project scope.

Master portfolio source — content should be changed only when Kevin intentionally updates the project/profile.